

EconomicBridge — Technical Specification

AI & Satellite Intelligence for Agriculture, Food Security & Aid Delivery · Operated by Bizra Farms Integrated Nigeria Limited

EconomicBridge is a deployed, multi-tenant satellite-intelligence platform serving West African governments, NGOs, and international bodies. It is designed for 52 administrative units (36 Nigerian states + FCT + 15 ECOWAS countries) with 10 pilot tenants live today, spanning over 700 LGAs/districts across Nigeria, Ghana and Senegal. Seven intelligence modules convert satellite and open statistical data into 24–72-hour early warnings and decision-grade reporting. The dashboard is open-access for situational awareness; institutional features are invite-gated per tenant.

Platform architecture

Five containerized microservices — Next.js 16 dashboard, FastAPI core API, satellite-ingestion service, ML inference service, and a notifications service — run on AWS ECS Fargate (eu-west-1) behind a path-routing Application Load Balancer. PostgreSQL 16 (RDS) holds a schema-per-tenant data model; ElastiCache Redis provides caching; model artifacts ship via S3; credentials live exclusively in AWS Secrets Manager. The entire stack — 140+ resources — is defined in Terraform, and CI/CD runs on GitHub Actions with OIDC federation (no long-lived cloud keys anywhere).

Intelligence modules

Module	Function	Primary inputs
Economic Visibility	Poverty & settlement mapping per LGA	WorldPop, nightlights, DHS-style indicators
Aid Coordination	Coverage gaps and duplication across agencies	HDX HAPI, partner records, bulk CSV
Farmland Protection	Encroachment & conflict early warning (24–72 h)	Sentinel-1 SAR, NASA FIRMS, ML conflict model
CropGuard	Crop-disease diagnosis + vegetation & price intelligence	Leaf imagery (ResNet-50), Sentinel-2 NDVI, market prices
ShockGuard	Flood / drought detection and alerting	Sentinel-1 SAR series, statistical detectors
Mobility Compass	Income & cost-of-living, displacement capacity	World Bank GNI/employment (USD-anchored, dual-currency)
SkillsBridge	Education access & connectivity targeting	UNICEF GIGA school locations, World Bank ICT

Live data feeds (running in production)

Source	Data	Cadence
Copernicus Sentinel-1 (CDSE)	SAR backscatter — all-weather flood/change signal	Weekly per tenant ROI
Copernicus Sentinel-2 (CDSE)	NDVI — crop vigor & anomaly	Weekly
NASA FIRMS	MODIS/VIIRS active-fire detections	Daily 06:00 UTC
N2YO	Live satellite pass tracking	Every 15 minutes
World Bank API	GNI/capita, employment (CC BY 4.0)	Monthly
UNICEF GIGA	School locations (100k+ for Nigeria)	Monthly
WorldPop	Gridded population	Weekly

Every datum carries provenance: dashboards badge LIVE observations distinctly from MODELLED baselines — the platform never presents synthetic data as measured.

AI / ML models

Model	Purpose	Status
ResNet-50 (12-class)	Crop-disease classification from leaf imagery; top-k output, confidence bands, human-review gating	Fine-tuned; 87.2% validation accuracy; serving in production

Model	Purpose	Status
Random Forest	Conflict prediction 24–72 h (lineage: the Citadel Kebbi State security platform)	Deployed pattern, retrained per pilot
Statistical detectors	Flood (SAR z-scores) and NDVI anomaly on live Sentinel series	Running on live Copernicus data

Security, tenancy & compliance

JWT authentication (15-minute access / 7-day revocable refresh, bcrypt password storage) with per-IP login rate-limiting. Strict multi-tenant isolation: schema-per-tenant in PostgreSQL plus request-level permitted-tenant enforcement — a state tenant can never read another state's data (HTTP 403), while accredited regional bodies (e.g., ECOWAS-type partners) hold read access across pilots. Personally identifiable data sits behind a signed Data-Processing-Agreement gate; the audit log is INSERT-only. Farmer inclusion is registration-free: phone numbers are sourced through partner agencies acting as data controllers (NDPA 2023 aligned), with multilingual SMS alerts (English, French, Portuguese live; Hausa, Yoruba, Igbo drafted pending native review).

Operations

Scheduled ingestion (8 jobs, manually fireable from the admin console), CloudWatch logging and alarms, automated monthly/quarterly PDF & CSV reporting via EventBridge Scheduler, and a mock-to-live gateway pattern for outbound channels (AWS SES email, AWS SNS SMS) so every integration can be exercised safely before carrier activation.

Live platform: <http://economicbridge-staging-alb-691775567.eu-west-1.elb.amazonaws.com> · Source: github.com/Abraz-babs/EconomicBridge_Sat_AI_ML · Contact: bizrafarms@gmail.com · +234 703 791 9465 · Birnin Kebbi, Kebbi State, Nigeria